IGNITION · WAYFARER · BRANCH CLAUDE/WAYFARER-ACCESSORIES · C3CB062 + VIEWER PASS · OWNER REVIEW · 2026-10-24

Accessories: owner review and handoff

The third prototype milestone (c3cb062) is accepted for review purposes only: not merged, no engineering gate released, not approved hardware. The register below separates visual choices the owner can approve now from engineering decisions that belong in the canonical Drive sheets. Every image is a PROTOTYPE review render; dimensions are measured readbacks of the as-built prototype, never ratings. The ship, its geometry and the accepted baselines are untouched.

DGT-017 HOLD · isolated studies | DGT-012 HOLD · GU-SIF proposed | FLIGHT_EVA HOLD · training simulation

ASSET-10 compact PLSS · choice DEFERRED | mass · power · payload · latch · life support · grip OPEN

1 Owner review and handoff register

Scope: the `claude/wayfarer-accessories` branch at c3cb062 (third prototype milestone) plus the viewer usability pass. Everything here is **PROTOTYPE**. Accepting it as a review milestone neither merges it into `main`, releases an engineering gate nor approves spacecraft hardware. Detail and evidence are in docs/ACCESSORIES.md, and the images are indexed in this folder's README.

The register has three parts. **A** lists visual choices the owner can accept or reject in chat now; they need no Drive work and change no engineering value. **B** lists engineering decisions that belong in the canonical Drive workbooks, where this repository only reads the owner rows. **C** lists ship-session items: this branch only requests them.

A. Visual decisions: approvable now

Accepting an item makes it the visual direction for that accessory (`L4_VISUAL_CANDIDATE` accepted for look). It doesn't replace an L2 proxy, settle a size or rate any part.

#	DECISION	EVIDENCE	NOTE
A1	GU drone family grammar: white vault, captive rail trucks, GU-SIF ends, GU stripe and plate	01, 02-10	the shape of GU-SIF is visual only; the standard is B2
A2	Functional identity of DR-EVA-01 (ORU guide frame, Brow inspection head), DR-EVA-02 (deflector, silvered faces, IR turret) and DR-EVA-03 (tool magazine, crew rail, tether anchor)	21, 05-07	sizes are unchanged and every body stays inside its envelope class; equipment ratings OPEN
A3	Docking cradle look on review stands (pedestal, V-guides, umbilical, lamp, orange release)	24	review fixtures, not ship stations; ratings are B2/B5
A4	Suit fidelity: boot shells and toe cap, knee and elbow covers, layer finishes (textile, hard shells, removable PLSS modules, grips), touch-point wear	20, 11, 12	wear is a soft vertex colour, not a texture
A5	PROTOTYPE lettering as placeholders: "GREAT UNION", the role line, "WF-Y155-EVA PROTOTYPE", "O2 / CO2 · SERVICE / COOL"	20, 11	text only, no emblem; owner Art / Fashion markings replace it
A6	Glove grasp as a display pose and its close-up view	22, 16	the middle of the handle stays hidden inside the glove; grip force and glove torque OPEN
A7	Viewer presentation: suit EVA actions playable, inherited crawl / climb / idle / walk / sit shown LOCKED and refused	23, 25	a labelling decision, not an approval of the suit's mobility; FLIGHT_EVA stays HOLD
A8	Viewer layout (grouped class names, action / status table, status card)	25, 18, 19	presentation only

B. Engineering decisions: canonical Drive work

Nothing on this branch changes these. Each one needs an owner row (or its answer) in the named workbook before any value moves into `data/` or a reference.

#	ITEM	STATE ON THE BRANCH	WHAT THE OWNER DECIDES	WHERE
B1	**DGT-017** L2 interface gate before detailed meshes	HOLD. The bodies are isolated studies; the ship keeps its L2 proxies	release the gate for one class at a time, after B3 and B5	S&E 54_DRONE_VALIDATION DGT-017

#	ITEM	STATE ON THE BRANCH	WHAT THE OWNER DECIDES	WHERE
B2	**DGT-012** GU-SIF / cartridge standard	proposed visual interface	approve or replace it. Needs: latch preload under 1.5 g thrust (F_axial = m × 1.5 g0, m `OPEN`), contact rating for HIGH_POWER charge, mate / demate life, keying, gloved manual-release force, power-loss state, tests (docs/ACCESSORIES.md "GU-SIF")	S&E 54 DGT-012 and DGT-SRC rows
B3	**Drone / placeholder clearance** (corrected)	stowed body over its L2 proxy in height: DR-EVA-01 **+0.12 m** (0.48 vs 0.36), DR-EVA-02 **+0.03 m** (0.39 vs 0.36), DR-EVA-03 **+0.04 m** (0.46 vs 0.42), DR-EVA-04 +0.11 m (0.35 vs 0.24). All inside the S&E 51 envelope classes; interior classes fit. The 042ee65 figures were 29 mm high (the stand's dock fitting was counted)	either grow the proxies after a placement and keep-out study or cap the body height; no automatic proxy change	S&E 51_DRONE_FLEET col G (classes); placement study by the ship session (C2)
B4	**Rail section**	the candidate truck needs Ø60 mm rail, a 32 mm web corridor to 75 mm below the rail centre and a Ø206 mm swept envelope (rail_section.svg)	rail section, web spacing, mount pattern; loads and ratings `OPEN`	owner interface row via the TOC (S&E 15_MODEL_INTERFACES); ship geometry by the ship session (C3)
B5	**Docking stations**	cradles exist only on review stands. Ship stations stay where the ship model puts them (DS-10 Brow docks, DS-11 SP-51 HOME docks, DS-13 spine track with operations `HOLD`, DS-14 aft cradles)	dock design: latch, charge and alignment ratings, cradle form at each DS position	S&E 53 (DOP-12 power `OPEN`) / 54 (DGT-012); positions by the ship session
B6	**Suit-reference readbacks**	torso width, depth, circle, access classes, sweep and EVA coverage are identical to the committed references. Moved: height 1.921 / 1.835 m (+3 / +2 mm, boot shell standoff); EVA_SUITED projected area max 1.280 / 1.182 m² (+0.002 / −0.001), min 0.473 / 0.439 m² (side rescue handles, 042ee65). Inherited crawl / climb coverage FAIL unchanged and not suppressed	accept the readbacks (then the owner's `workflow.py eva` step refreshes the references) or reject the boot standoff; the projected area feeds the thermal case	S&E 44_EVA_SUIT_BLENDER (ASSET-08 boots), 45_EVA_ENGINEERING_CASES H-06
B7	Compact PLSS (ASSET-10)	`parked`. Baseline pack unchanged; every subsystem size is `OPEN`; the 0.90 m ring-B port is not an owner-required suited route (EVA-040)	nothing until EVA-051's calculation releases masses, capacities and energy	S&E 43 EVA-051, 44 ASSET-10
B8	All physics	`OPEN`: drone mass, power, speed, reach, payload, latch strength, thermal endurance; suit life support, glove torque, grip	values in owner rows; until then the tests keep them `OPEN`	S&E 51 / 53 / 54, 43 / 45
B9	FLIGHT_EVA	`HOLD`; every EVA scene is a training simulation	not opened by an animation or collision check	S&E 43 / 44 QA

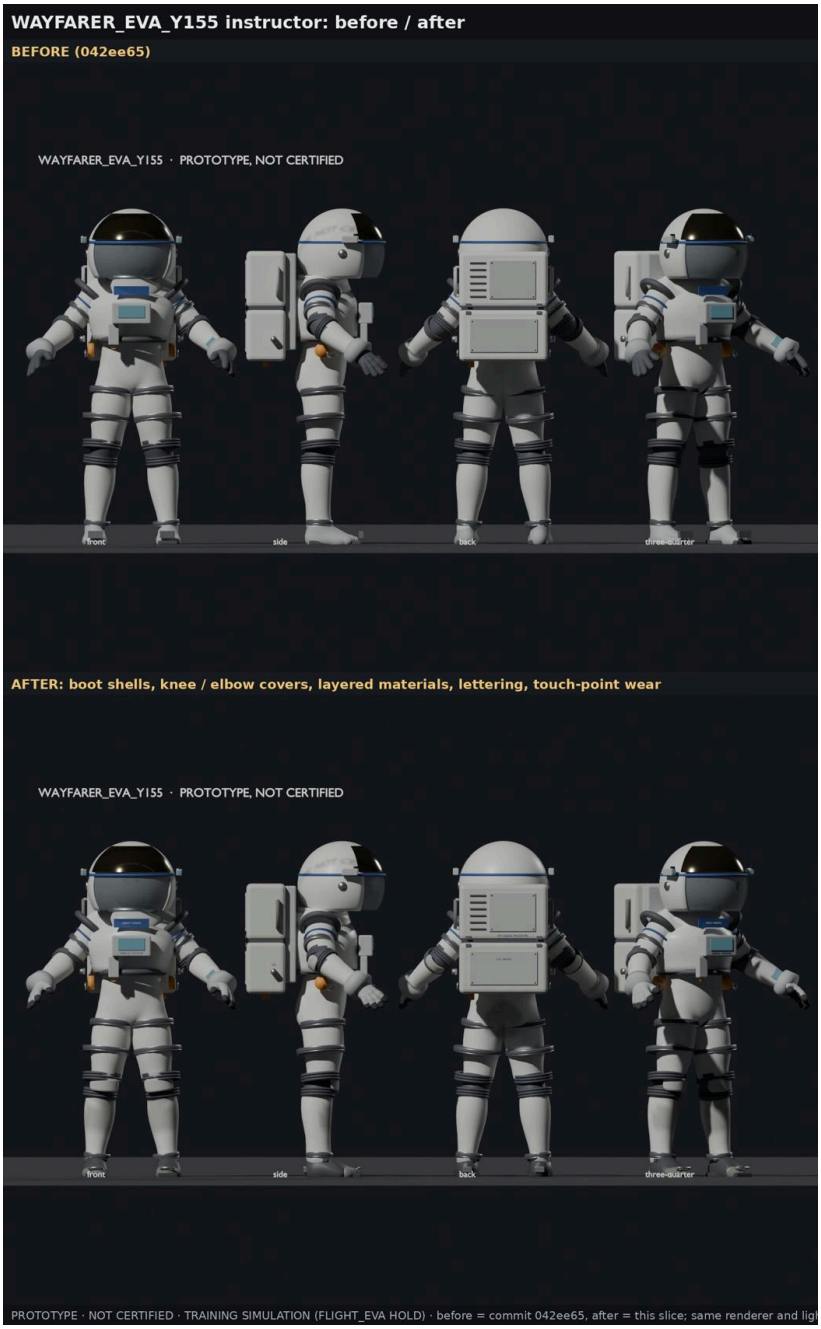
C. Ship-session handoff (requests only; the ship session owns the geometry)

#	REQUEST	FROM THIS BRANCH
C1	habitat rover deck-guide path near the DS-05/07 floor cassettes (the 40 mm review channel is provisional)	17
C2	external drone placement and keep-out study with the corrected heights (B3)	01, 21, B3 table
C3	rail section, web spacing and mount pattern from the ship side (B4)	rail_section.svg
C4	source map: add S&E `56_DRONE_TECH_ANCESTRY!A1:I9` next to `[sources.drone_fleet]` (shared source routing)	docs/ACCESSORIES.md request 4
C5	integration of an accepted class (after B1): its GLB, export bindings and dock anchors	not started; this branch edits no ship file

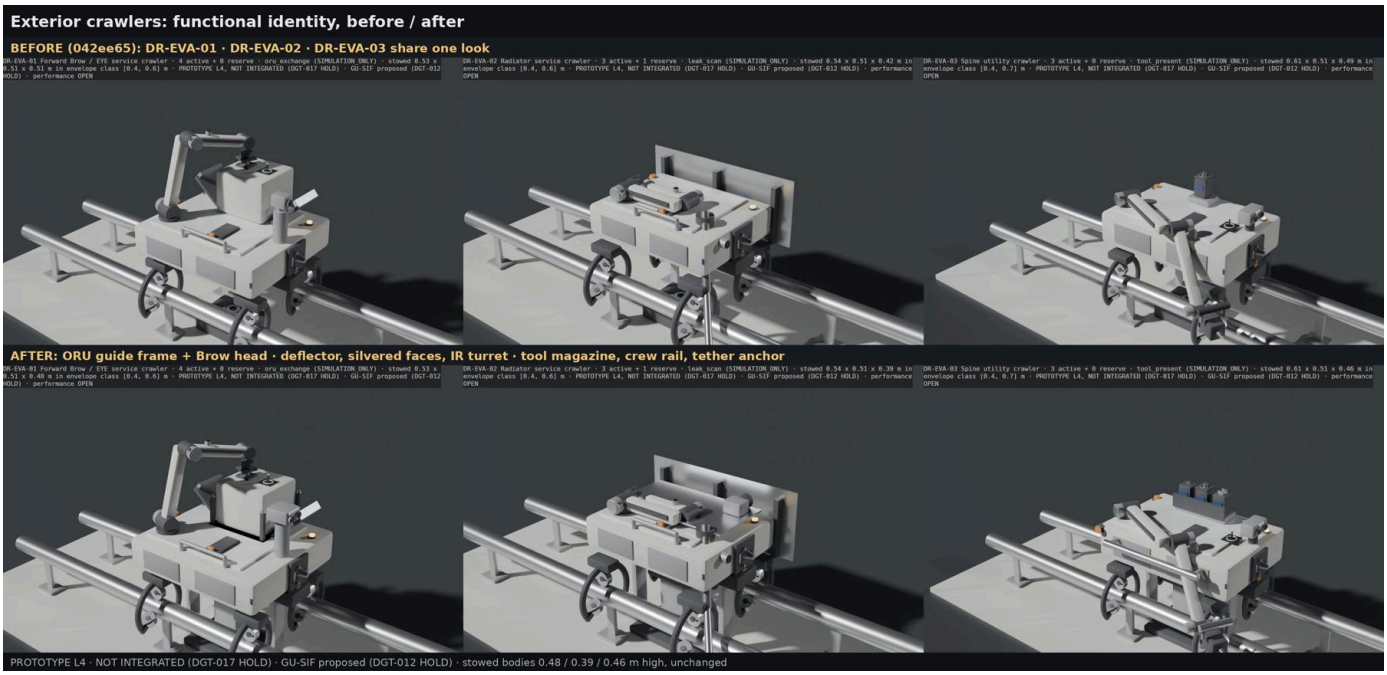
Branch state: isolated and not rebased or merged. No accepted reference, shared parameter, registry or ship-runtime file was edited. At c3cb062 the scoped quick pipeline reported 0 geometry changes (`runs/20261024T132405Z-quick`, kept as evidence).

Evidence numbers refer to the images in `accessories/review/` (README index); the register itself is `accessories/review/HANDOFF.md`.

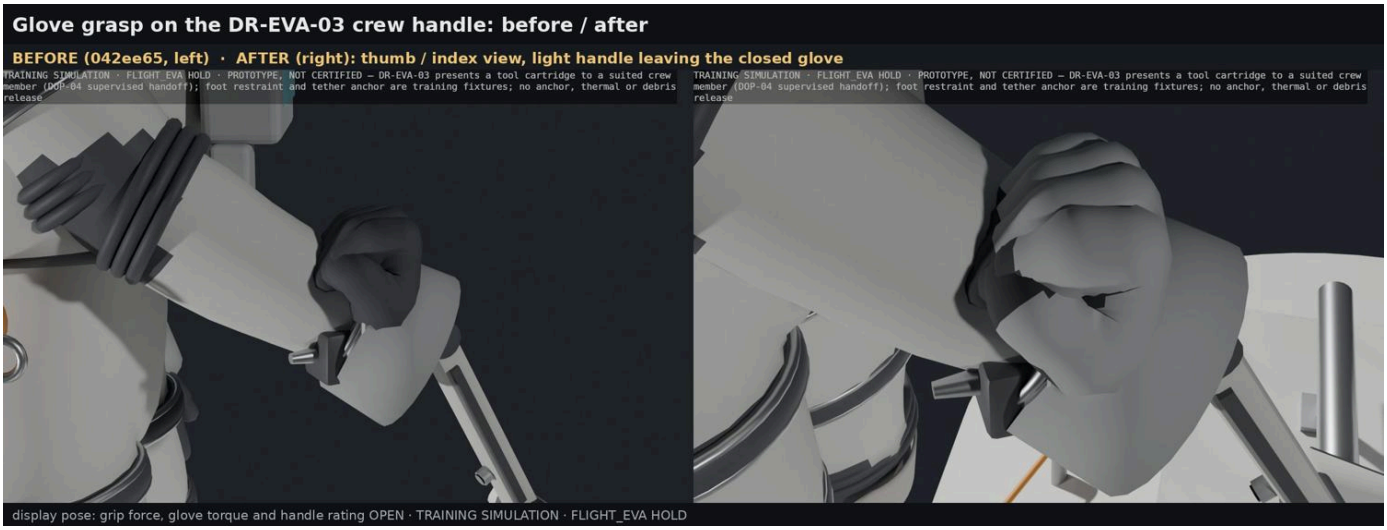
2 Before and after



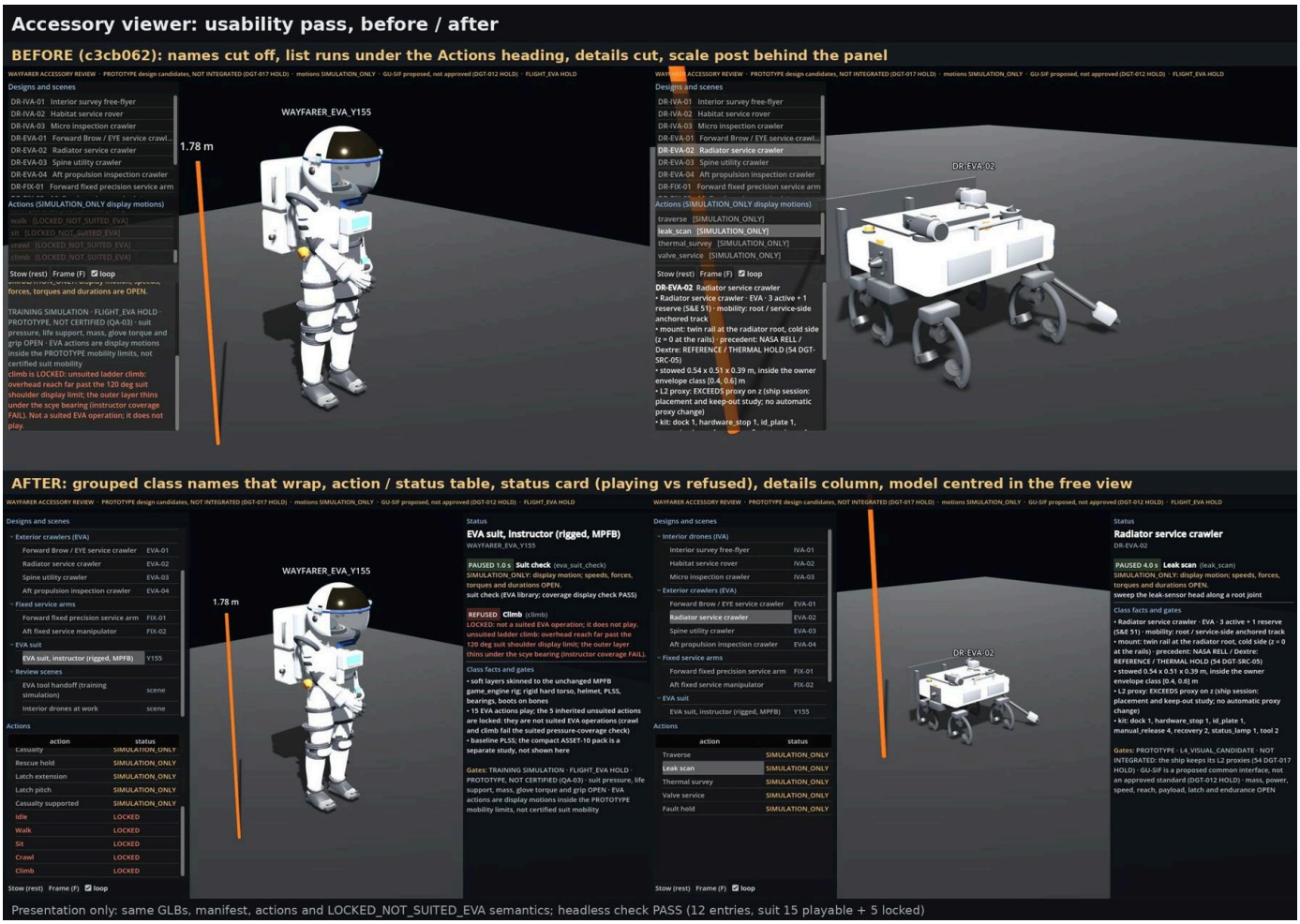
Instructor suit, 042ee65 build (top) and now (bottom), same renderer and lights.



DR-EVA-01 / 02 / 03 before and after the functional identity pass.



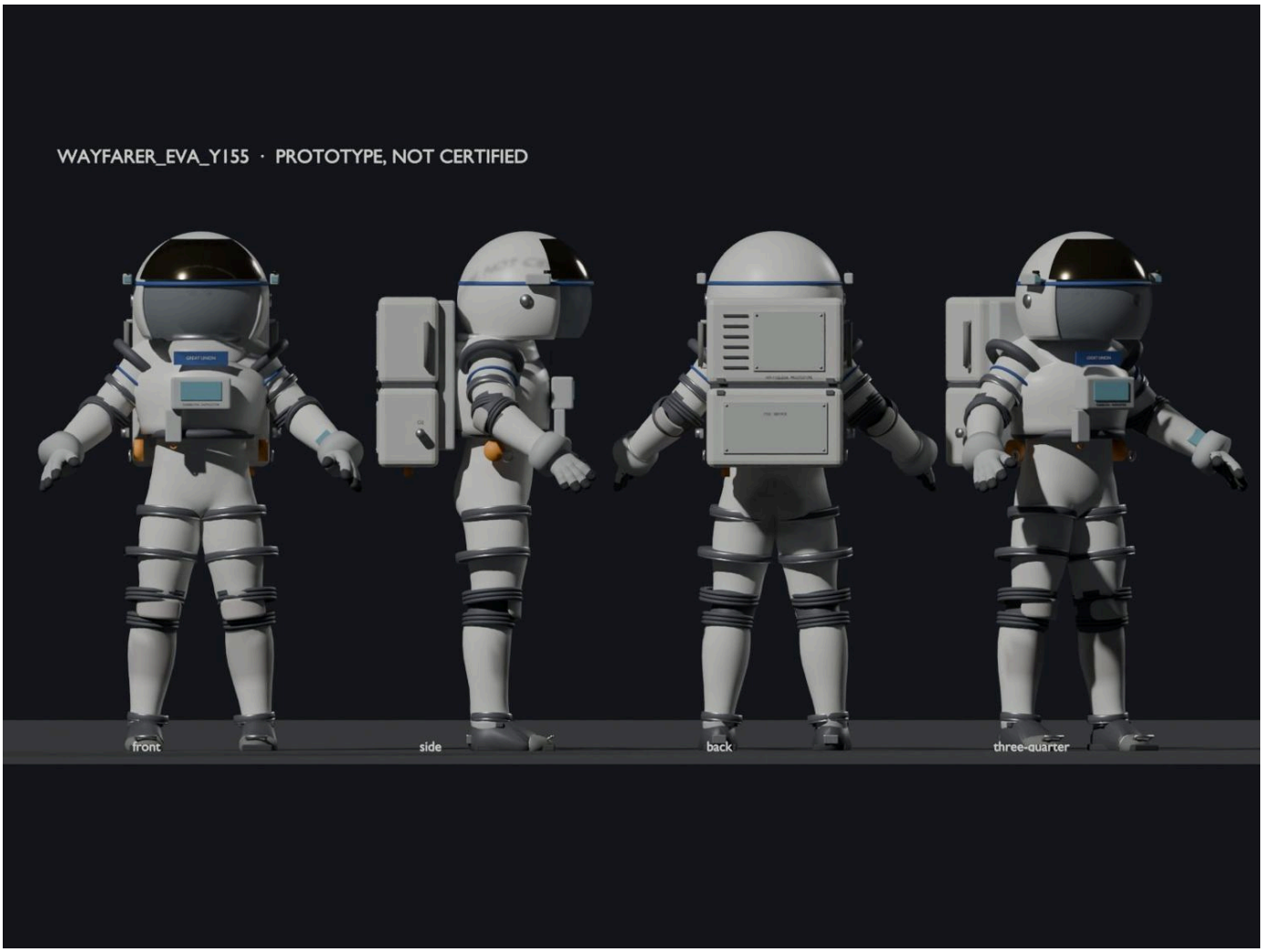
Grasp close-up before and after: the handle now leaves the closed glove between thumb and index finger in view.



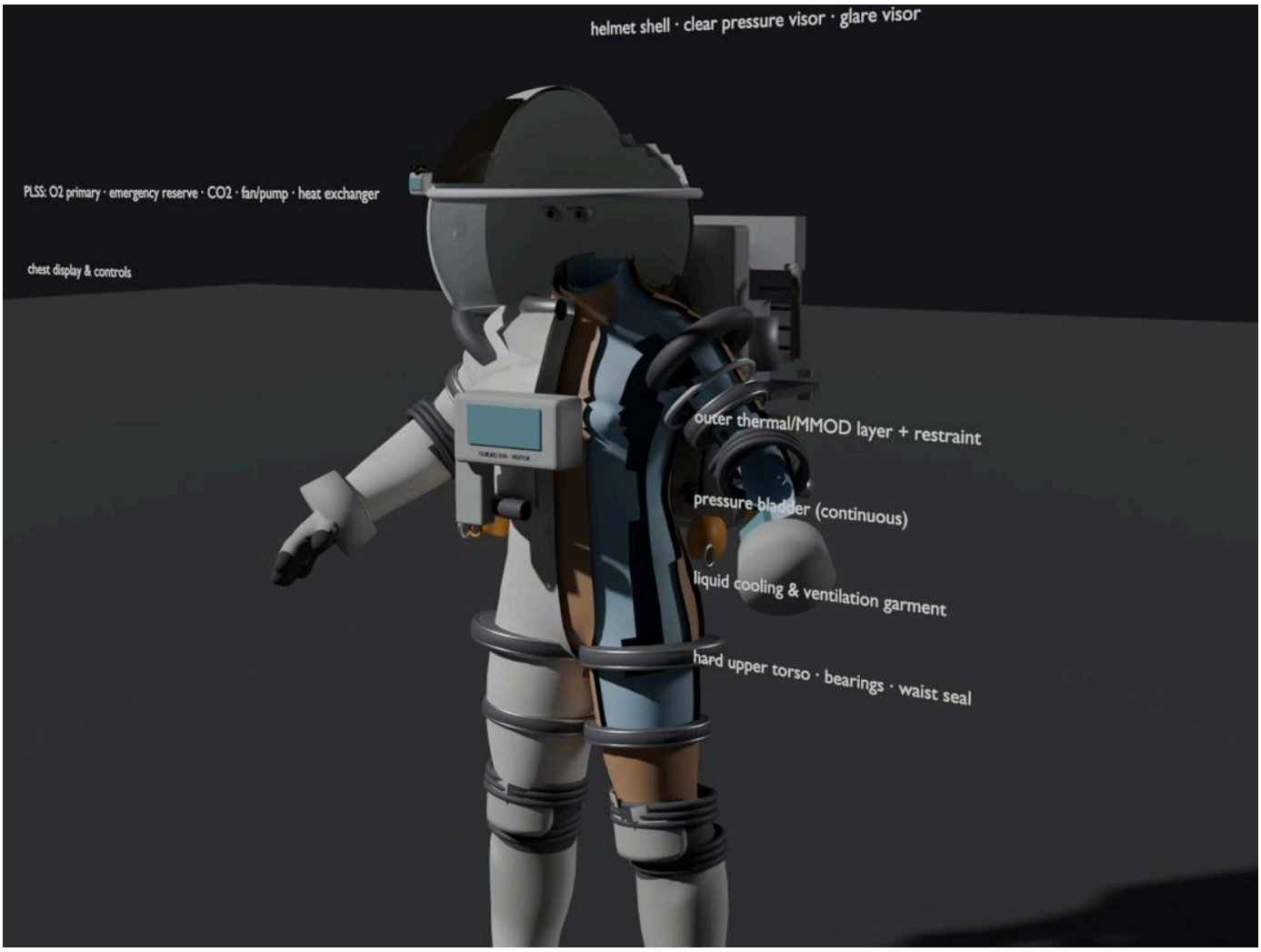
Accessory viewer before (c3cb062) and after the usability pass: grouped class names, action / status table, status card with PLAYING / REFUSED, details column.

3 Suit: what changed

- **Boots:** rigid heel-and-arch shell and toe cap over the soft layer, black rubber service sole, heel fitting, instep strap, and a toe-hook bar that slides under the restraint's toe bail. The review foot restraint gained toe bails and heel clips.
- **Joint covers:** replaceable knee and elbow pads over the convolute gaps, sized inside the convolute rings so they add no extent.
- **Layers:** matte woven textile (outer) / smoother hard composite (hard torso, helmet) / warm grey removable equipment (PLSS modules) / dark elastomer joints. Gloves grey with black rubber palm and fingertip grips.
- **Markings and wear:** text only, no emblem: GREAT UNION plate, GUIDED EVA role on the chest module, a PROTOTYPE serial field, O2 / CO2 / COOL service labels; soft touch-point wear at knees, shins, elbows and seat.
- **Kept:** the MPFB rig and its actions, the baseline PLSS, torso width, depth, circle and every access class.



Turnaround: front, side, back, three-quarter.



Cutaway: layers, hard torso, PLSS internals (display layout, not packaging).

4 Measured readbacks (instructor / newcomer)

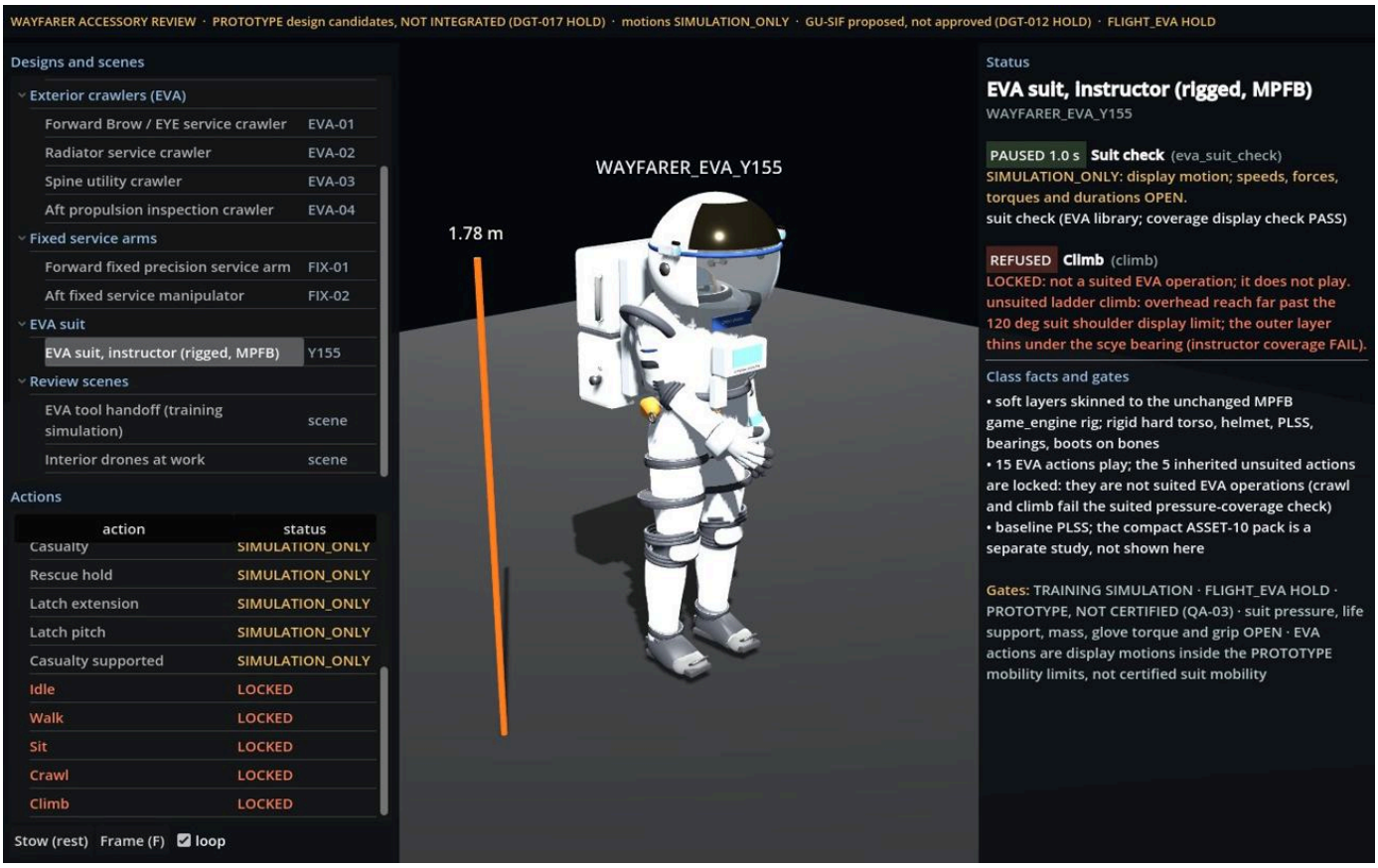
READBACK	COMMITTED REFERENCE	BASELINE NOW	COMPACT STUDY
Pressure coverage, 15 EVA actions	PASS	PASS	PASS
Inherited unsuited `crawl` (both) and `climb` (instructor)	FAIL	FAIL, NOT_SUITED_EVA	FAIL, NOT_SUITED_EVA
Torso width / depth with PLSS	0.640 / 0.598 · 0.798 / 0.769 m	identical	0.664 / 0.664 · 0.718 / 0.689 m
Suited circle	0.909 / 0.885 m	identical	0.890 / 0.867 m
Access classes EVA_SUITED / OPERATIONAL / UNSUITED	0.857 / 1.074 / 0.658 0.832 / 1.026 / 0.602	identical	0.815 / 1.020 / 0.658 0.792 / 0.974 / 0.602
Height	1.918 / 1.833 m	1.921 / 1.835 m	1.921 / 1.835 m
Projected area min / max (thermal readback)	0.467 / 1.278 · 0.432 / 1.183 m²	0.473 / 1.280 · 0.439 / 1.182 m²	0.445 / 1.248 · 0.410 / 1.149 m²

Height (+3 / +2 mm) and the area maximum moved with the boot shells' 2.5 mm standoff; the minimum moved with the rescue handles last slice. Readbacks for the owner's thermal model, not approval. Committed references not refreshed.

5 Validation semantics

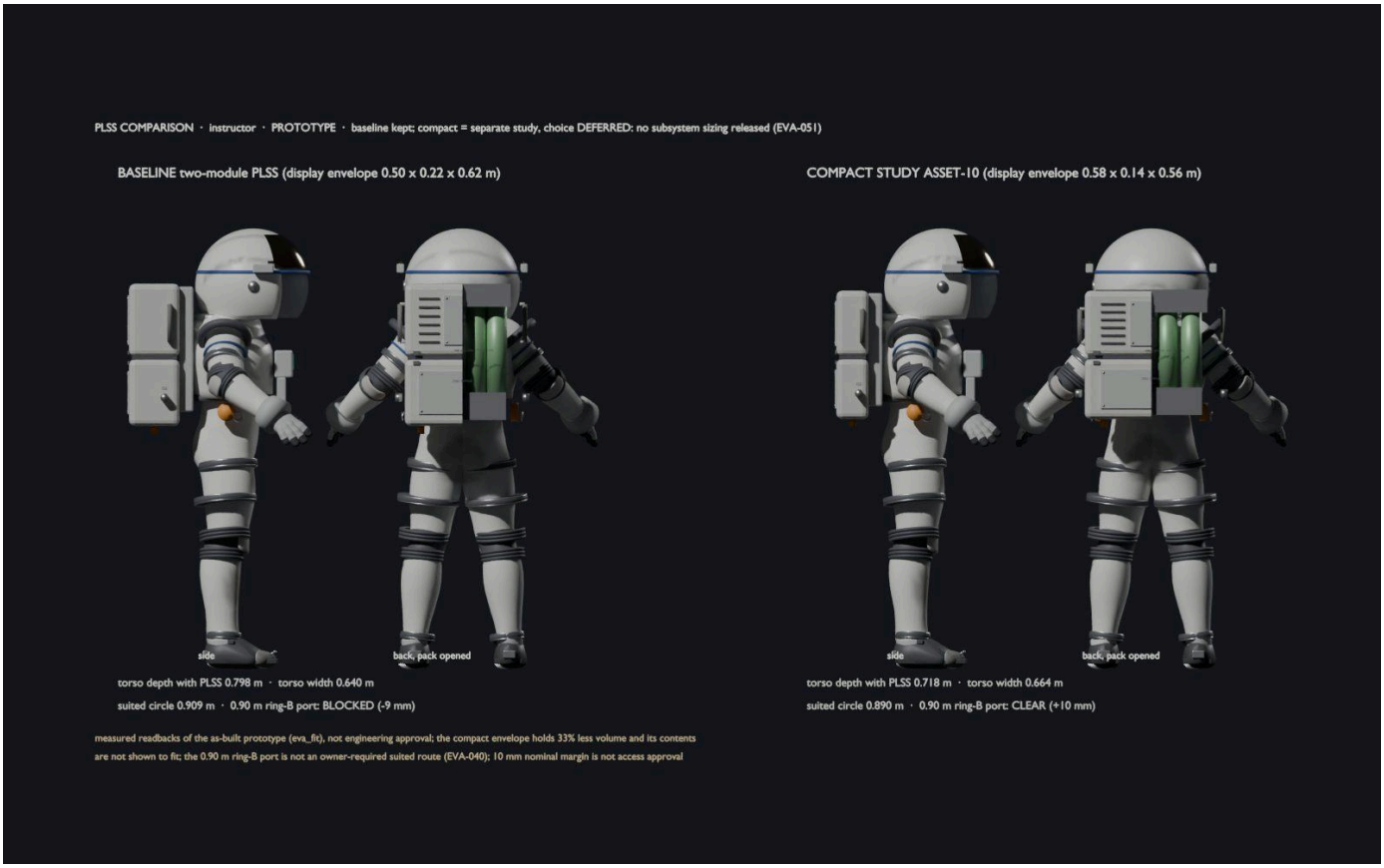
All 15 EVA actions pass coverage. The inherited `crawl` and `climb` come from the unsuited character rig and are not suited EVA operations; they now cannot be read or played as one: listed in `SHARED_ACTIONS` (kept disjoint from the EVA list by a test), tagged `NOT_SUITED_EVA` in the suit asset, reported with `use` and `eva_coverage_pass` beside the unchanged overall `coverage_pass = false`, and locked in the viewer.

Climb investigation: 2-3 vertices per arm at the top of the upper arm, just outside the scye bearing, up to 3.8 mm frame by frame (2.5 mm at the check's step), during each overhead reach. The climb lifts the arm about 176° from the trunk against the suit's 120° limit (EVA actions reach 116-121°): an unsuited motion outside suit mobility, not a suit-geometry defect. Nothing was relaxed to hide it.



Accessory viewer: the suit plays its 15 EVA actions; the inherited unsuited actions are listed LOCKED and refuse to play.

6 Compact PLSS: packaging feasibility

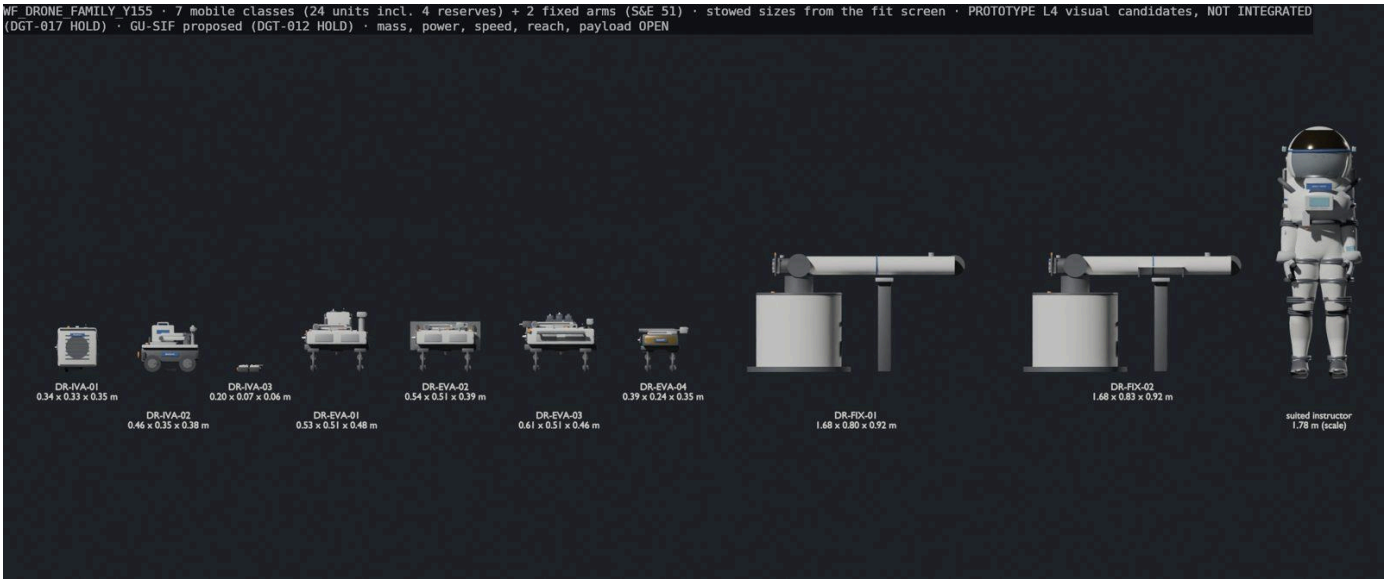


Baseline pack (kept) and the compact ASSET-10 study, packs opened; readbacks labelled.

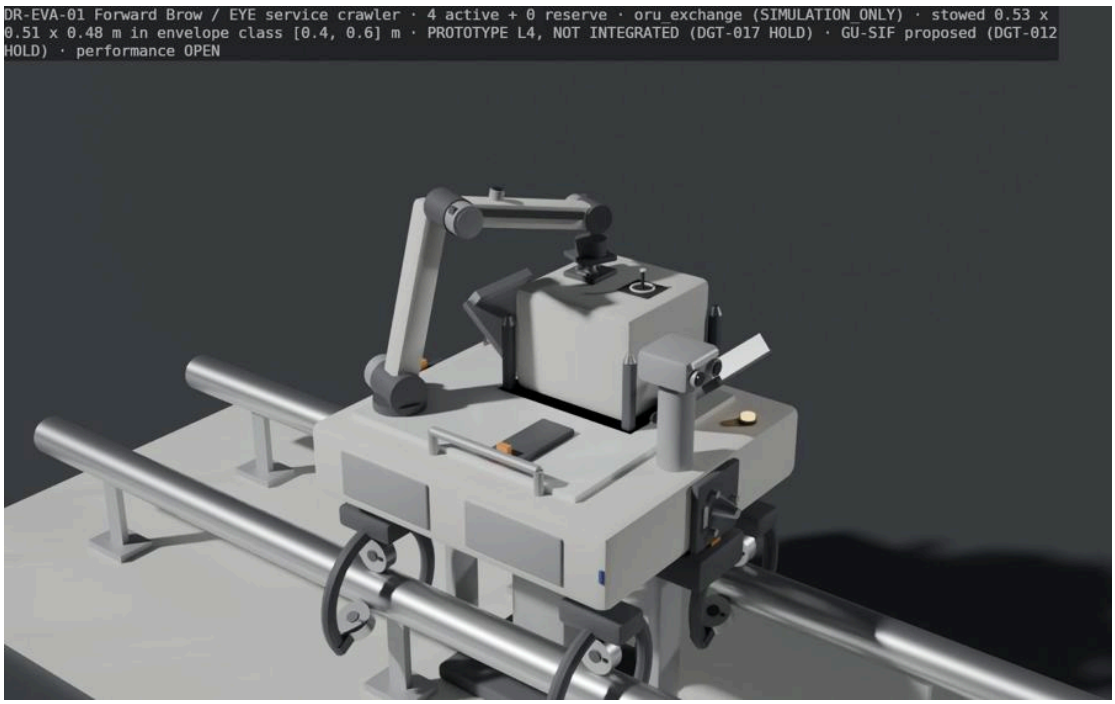
No owner row (S&E 43/44, Technology 53) gives a dimension, volume or mass for any PLSS subsystem; EVA-050 keeps the 0.50 × 0.22 × 0.62 m pack only as a labelled packaging case and EVA-051 commissions the sizing calculation. For oxygen storage, emergency oxygen, CO_2 removal, ventilation, cooling, battery, communications, plumbing, structural mounting and maintenance access the owner-specified dimension is **none** and the sizing data are OPEN. The compact envelope holds a third less volume; its only measured gain is at the 0.90 m ring-B port, which EVA-040 says suited transit does not need. **Decision: deferred.** Full matrix in `docs/ACCESSORIES.md`.

7 Drone family

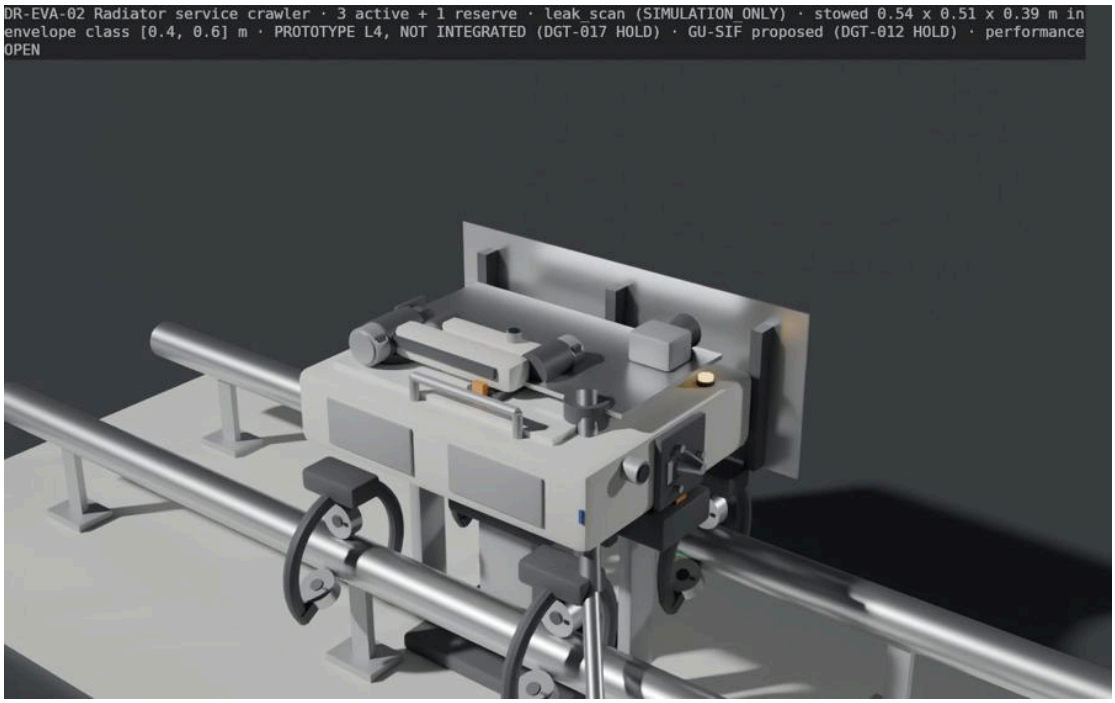
DR-EVA-01 gains an ORU guide frame and a Brow inspection head; DR-EVA-02 an upper deflector, silvered blade-side faces and an IR radiometer turret (new `thermal_survey` action); DR-EVA-03 a three-slot tool magazine, a crew handrail and a tether anchor. Review stands carry PROTOTYPE docking cradles (pedestal to the underside GU-SIF-M, V-guides, umbilical, lamp, manual release); not on the ship. No stowed height grew. Correction: the 042ee65 heights counted the stand's dock fitting as body; the bodies are 0.48 / 0.39 / 0.46 m, so they exceed the ship placeholders by +0.12 / +0.03 / +0.04 m (not +0.15 / +0.06 / +0.07).



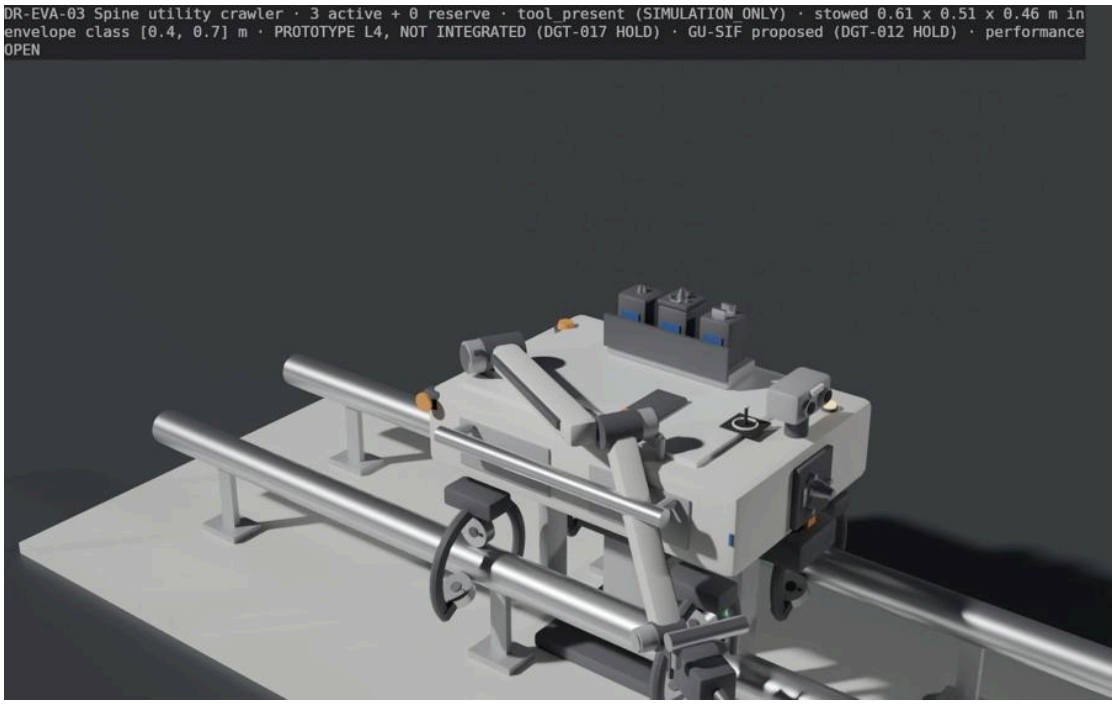
All nine designs stowed at one scale with the suited instructor (1.78 m).



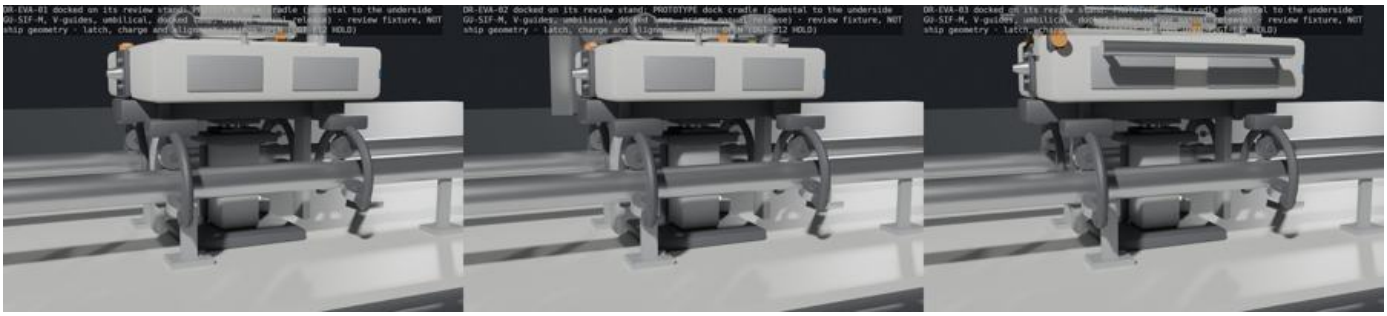
DR-EVA-01 Brow / EYE crawler · oru_exchange · 0.53 × 0.51 × 0.48 m



DR-EVA-02 radiator crawler · leak_scan · 0.54 × 0.51 × 0.39 m



DR-EVA-03 spine crawler · tool_present · 0.61 × 0.51 × 0.46 m



DR-EVA-01 / 02 / 03 docked on their review stands: PROTOTYPE docking cradles (review fixtures, not ship geometry).

8 Still HOLD or OPEN

- DGT-017 (bodies isolated, ship keeps L2 proxies), DGT-012 (GU-SIF proposed), FLIGHT_EVA (training simulation), DGT-007 (radiator view factor and cooldown).
- Drone mass, power, speed, payload, reach, manipulator load, latch strength, thermal endurance; dock cradle latch, charge and alignment ratings.
- Suit life support (O_2, CO_2, heat rejection, duration), pressure setpoint, glove torque and grip, restraint and handrail loads; ASSET-10 packaging.
- Ship session owns drone placement and keep-out studies, rail interfaces and the 56_DRONE_TECH_ANCESTRY source-map entry.

Generated from the accessory branch (`eva_preview.py`, `drone_preview.py`, `accessory_scene.py`, `tools/accessory_viewer.py`); images also committed under `accessories/review/`; notes in `docs/ACCESSORIES.md`.